

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories. The analysis moves beyond descriptive summary statistics to test specific business questions in SQL and Power BI, with the goal of finding not just what the numbers are, but what they imply for subscription growth, discount policy, and customer segmentation. The most significant result is a complete absence of female subscribers in the dataset, detailed in Section 5.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in the Review Rating column

3. Exploratory Data Analysis using Python

Data preparation and cleaning was done in Python before any SQL or dashboard work:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	0
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake_case for better readability and documentation.
- **Feature Engineering:**
 - Created age_group column by binning customer ages.
 - Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in PostgreSQL to answer ten business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Male customers generate \$157,890 vs. \$75,191 for female customers roughly double, which tracks closely with the 68/32 split in customer counts rather than a difference in individual spending.

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:00.133

839 customers used a discount and still spent above the \$59.76 average roughly 21% of all customers, showing discounts are not just attracting price-sensitive, low-spend shoppers.

3. Top 5 Products by Rating – Found the products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Gloves (3.86), Sandals (3.84), and Boots (3.82) lead all accessories or footwear, not Clothing, despite Clothing driving the most revenue.

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Express shipping customers spend \$2.02 more on average (\$60.48 vs. \$58.46) a small but consistent premium, suggesting faster shipping correlates with slightly higher-intent purchases.

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Subscribers spend almost the same per order as non-subscribers (\$59.49 vs. \$59.87) the subscription itself is not associated with a meaningfully larger basket.

6. Discount-Dependent Products – Identified the 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Hats lead at a 50% discount rate, followed closely by Sneakers, Coats, Sweaters, and Pants all mid-to-high price items, suggesting discounting is concentrated on bigger-ticket goods.

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

79.9% of customers (3,116) qualify as Loyal (11+ prior orders), with only 2.1% still New this is a highly retained customer base already, which is part of why Section 5's finding matters.

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Jewelry, Blouses, and Sandals/Jackets lead their respective categories, each with 145–171 orders demand is fairly evenly spread across each category's top items rather than concentrated in one.

9. Repeat Buyers & Subscriptions – Checked whether customers with more than 5 previous purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Only 958 of 3,476 repeat buyers (27.6%) are subscribed almost identical to the 27.0% subscription rate across all customers. Repeat purchasing does not meaningfully predict subscription.

10. Revenue by Age Group – Calculated the total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Revenue is nearly flat across age groups (\$55.8K–\$62.1K) no single age segment stands out as a growth lever, unlike the subscription split by gender below.

5. Key Finding: Subscription Access by Gender

Cross-referencing subscription status against gender surfaces the most consequential result in this analysis, one not captured by any single query above: **zero of the 1,248 female customers in this dataset are subscribed**, compared to 1,053 of 2,652 male customers (39.7%). This is not a disparity of degree, it is a complete absence.

This gap cannot be explained by differences in engagement or satisfaction: male and female customers spend almost identically (\$59.54 vs. \$60.25 average order) and rate purchases almost identically (3.75 vs. 3.74 average review). The only place the two groups diverge completely is subscription status. A finding this clean — a hard 0%, not a skew — should be verified with the data or product owner before being reported as a business insight: it may indicate the subscription program was never extended to female customers, a collection artifact in this particular extract, or gender acting as an undocumented eligibility filter.

6. Dashboard in Power BI

An interactive Power BI dashboard was built to surface these findings visually. It includes seven filters (Subscription, Gender, Discount Applied, Category, Age Group, Shipping Type, Payment Method) that cross-filter every visual on the page, four headline KPIs (total revenue, customer count, average purchase amount, average review rating), and six charts: Revenue by Gender, Revenue by Category, Revenue by Age Group, Subscription by Gender, and Top 3 Products by Review Rating (with sample size labeled per product to avoid misleading small-sample ratings). A narrative panel states the subscription finding directly rather than leaving it to be inferred from the charts alone.



The Subscription by Gender chart (bottom-left) is the dashboard's central visual: the Male bar splits 40%/60% between subscribed and not, while the Female bar is 100% “No,” with no subscribed segment at all — making the Key Finding from Section 5 visible at a glance.

7. Business Recommendations

- **Audit subscription eligibility by gender.** Before running any subscription campaign, confirm whether the program has ever been offered to female customers. A 0% rate with no difference in spend or satisfaction points to an access issue, not a demand issue.
- **Decouple loyalty programs from subscription marketing.** Repeat-purchase behavior (27.6% subscribed) tracks the overall average (27.0%) almost exactly rewarding loyal customers will not, by itself, convert them into subscribers.

- **Re-test discount strategy around basket size, not just conversion.** Discounted orders average \$59.28 vs. \$60.13 without a discount discounts are not lifting order value. A spend-threshold promo (e.g., “\$75+”) is worth testing in place of flat codes.
- **Prioritize Clothing and Accessories in campaigns.** The two categories already carry 77% of total revenue; product marketing spend is likely better concentrated here than spread evenly across all four categories.
- **Don't target by age group for revenue growth.** Revenue is flat across all four age bands (\$55.8K–\$62.1K) unlike the subscription gap, age is not a lever worth prioritizing.